



Software Engineering

04.05.2025

—

Final Report

HealthHubSystem

Group Members Info

Name	Roll Number	Section
Anas Imran	21i-2520	C
Huzaifa Khalid	21i-0527	C
Nida Azam	21i-0433	C

Roles	
Role	Member Name
Project Owner	Nida
Scrum Master	Anas
Developer	Huzaifa

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Introduction:

In our project, Health Hub, developed using the MERN Stack, we've crafted a transformative healthcare platform catering to three distinct user roles: administrators, doctors, and clients. Administrators oversee the platform's operations, managing user accounts and ensuring system functionality. Doctors utilize Health Hub for appointments, accessing patient records securely, and providing personalized care. Clients benefit from personalized health assessments, appointments with doctors, and streamlined management of medical records and medications. With a focus on user-centric design and advanced technology, Health Hub empowers administrators, doctors, and clients alike to engage with healthcare services efficiently and securely, fostering a collaborative approach to wellness management.

Project Rational:

The rationale behind the HealthHub project lies in addressing the existing gap in comprehensive healthcare management solutions. Presently, there's a scarcity of smart applications catering to the needs of hospital staff, including administrators, doctors, and clients. Health Hub aims to fill this void by providing a versatile platform where users can effortlessly create accounts and access a wide array of healthcare services tailored to their requirements.

For clients, Health Hub offers features such as predictive disease diagnosis based on symptoms and the flexibility to schedule appointments, both online and onsite, promoting convenience and proactive health management. Doctors can efficiently manage their appointments and conduct consultations online, thereby optimizing their workflow and broadening access to healthcare services. Administrators play a pivotal role in overseeing system operations and managing user accounts across different roles within the hospital ecosystem. By offering functionalities tailored to the diverse needs of hospital staff and clients, Health Hub seeks to streamline healthcare delivery and enhance overall efficiency.

To achieve these goals, the project is structured as a web-based application with a 3-tier layered architecture, leveraging popular web technologies. Adopting the agile methodology, particularly SCRUM, ensures adaptability and responsiveness to evolving requirements. The project consists of 25 user stories divided into three sprints, each meticulously planned and evaluated to monitor progress

effectively. Through this approach, Health Hub endeavors to revolutionize healthcare management and elevate the healthcare experience for both providers and clients.

Project Architecture:

HealthHub project implements a three-tier layered architecture, featuring the presentation layer, business logic layer, and database layer. At the presentation layer, users interact with the application interface

developed using HTML, CSS, JavaScript, React, and Material UI. This combination ensures a visually appealing and user-friendly interface that enhances the overall user experience.

The business logic layer, implemented using Node.js, encapsulates the application's functionality, managing various processes and features to meet user requirements efficiently. Node.js facilitates the seamless integration of server-side logic, enabling robustness, scalable application operations and at the same time keeping in view the privacy and security of data.

For data storage and management, the database layer leverages MongoDB, providing a reliable and scalable solution. MongoDB's flexible document-based model enables efficient data storage and retrieval, ensuring the integrity and security of user data within the system.

The adoption of this three-tier architecture, combined with technologies like React and Material UI, offers numerous advantages. It simplifies code organization, enhances maintainability, and facilitates collaboration among development teams. Moreover, React and Material UI contribute to the creation of a visually appealing and intuitive user interface, enhancing user engagement and satisfaction.

Product Backlog:

ID	User Story	Priority	Acceptance Criteria
1	As a user, I want to register on Health Hub so that I can access and view my health analytics, enabling me to make informed decisions and maintain my well-being effectively	1	Given that I am on the HealthHub registration page, when I fill in the required registration fields with valid information including my full name, email address, and password, then upon submission, a new user account should be created for me.
2	As a user of Health Hub, I want to log in with my specified role and password so that I can access personalized features and functionalities based on my role within the healthcare ecosystem.	1	Given that I am on the HealthHub login page, when I enter my registered email address and password corresponding to my specified role (e.g., administrator, doctor, or client), then upon submission, I should be authenticated and granted access to the system.
3	As a user I want to view my health analytics so that I can take rational choices and maintain my well-being	2	Given that I have opened the health assessment menu when I click on the view health analytics then it should display all the health-related options in human readable way.
4	As a user I want to receive my health status and risk factor so that I can keep record of my health stats and devise a plan accordingly.	2	- Given that I have opened the health analytics option when I fill in the required data then it should display all the stats and risk factor in a more visualized way. - Given that I have opened the health analytics option when I fill the required data wrong then it should display error message and ask me to re-enter the information.
5	As a user I want to get recommendations from the system according to my health stats so that I can get a better health plan.	3	Given that it displayed all the health analytics options when I ask for recommendation then it should perform certain function on my data and provide me with personalized proactive recommendations.
6	As a user, I want to access health care professionals through telehealth so that I do not have to go outside for booking appointment.	2	Given that I have selected for telehealth feature when I search for professional then it displays all the health care professionals in my area with their contact information
7	As a user, I want to schedule virtual consultations with healthcare professionals through HealthHub's telehealth feature to obtain medical advice and treatment	3	Given that users are presented with options to schedule virtual consultations with healthcare professionals when accessing the

	conveniently.		schedule virtual consultations then it should book appointments with the professional. Given that user has accessed the schedule virtual consultation when selected a desired time slot, the system confirms the appointment and notifies the user of the scheduled virtual consultation
8	As a user, I want to securely store my medical records within Health Hub so that I can maintain a centralized repository of my health information at any time.	5	Given that users have medical records when user click on view records then users can view and access their stored medical records from any device with internet access, ensuring seamless accessibility.
9	As a user, I want to access my test results in HealthHub so that I can compare my past medical test history and evaluate the changes in result.	5	Given that users have accessed test records when user click on view test record history then user can search tests using keywords or filters.
10	As a user, I want to access treatment plans stored in HealthHub so that I can review my past track of my ongoing healthcare journey and devise a new plan based on that.	4	Given that users have accessed treatment plans when user click on Give recommendation then user can get new treatment plan based on past plans.
11	As a user, I want to share specific medical records with my healthcare providers through HealthHub so that to facilitate continuity of care and informed decision-making	6	Given that the user has medical history when user wants to share information then User can select and share individual medical records or sets of records with designated healthcare providers securely
12	As a user concerned about my long-term health, I want HealthHub to utilize modern techniques to predict potential health issues early, allowing for timely interventions and preventive measures.	5	Given that I am a user of HealthHub, when I regularly input my health data and engage with the platform's features, Then the platform should use modern techniques to analyze my data over time, identifying any emerging patterns or trends indicative of potential health issues
13	As a user of HealthHub, I want a platform which analyzes my provided health data so that it provides me with personalized insights into my health risks	5	Given that I am a user of HealthHub, when I provide my health data such as medical history, lifestyle habits, and biometric measurements then the platform should use advanced technologies to analyze my data and identify potential health risks.

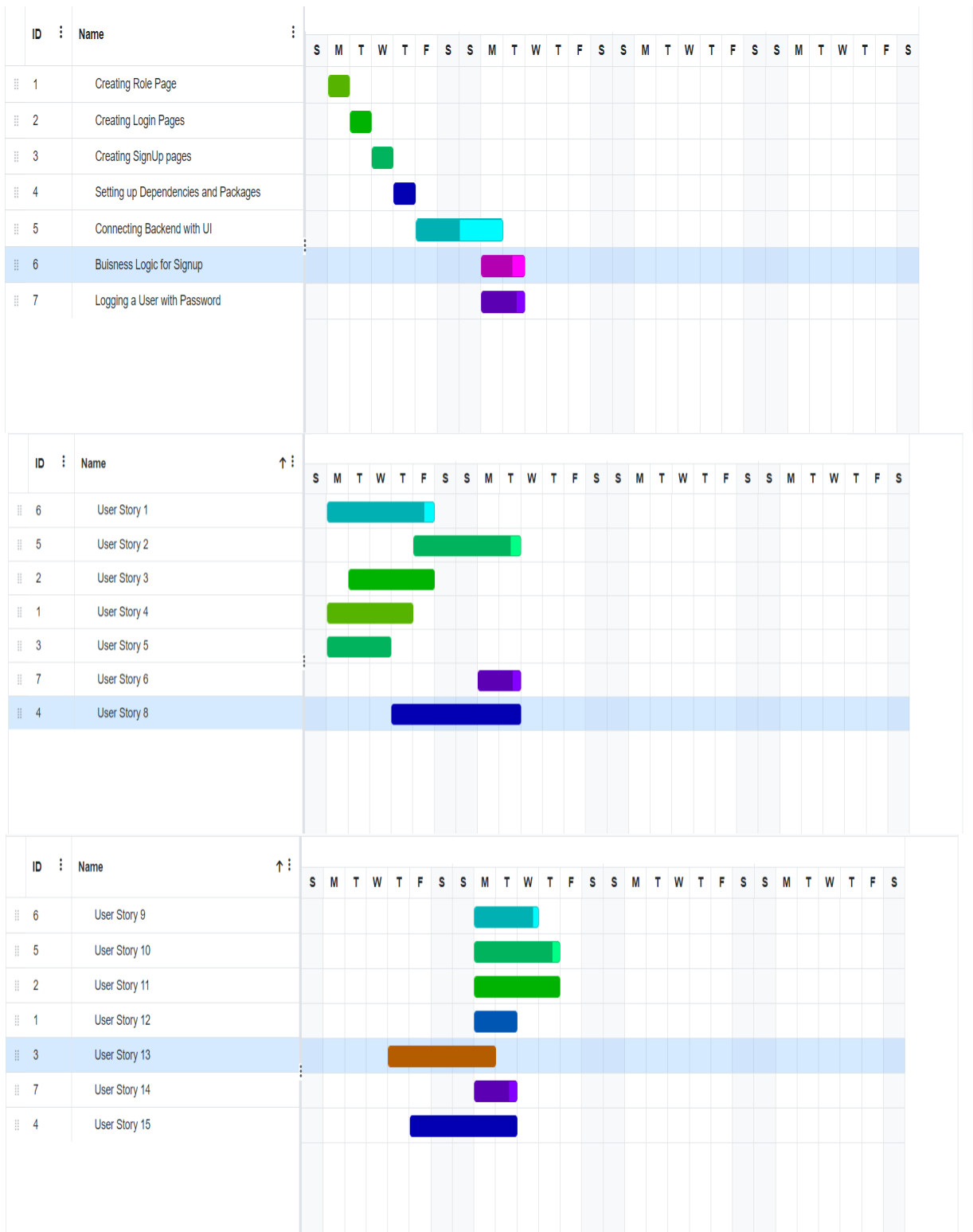
14	As a user of HealthHub, I want the platform to utilize advanced technologies to provide tailored health recommendations based on predictive health analytics.	6	Given that I am a HealthHub user, when I input my health data, including medical history, lifestyle habits, and biometric measurements, then the platform should employ
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			contemporary methods to analyze my data and anticipate potential health risks.
15	As a busy professional, I want quick access to health education and resources for stress management techniques so that I can manage my stress level and improve my overall wellbeing.	8	Given Health hub is accessible to user when user navigate to main menu then there is appropriate section of “Stress Management” and clicking to this section provide valuable resources to stress management in form of videos or articles
16	As a user concerned with managing my blood pressure effectively, I want comprehensive knowledge on managing sudden rising and falling of blood pressure in form of articles, videos, and pictures so that I can reduce risk of blood pressure related issues in future	8	Given that I am a user concerned with managing my blood pressure effectively when I access the HealthHub platform then this should prominently address the sudden rising and falling of blood pressure and should contain a variety of resources including articles, videos, and pictures.
17	As a user facing heart disease issues, I want to gain comprehensive information and techniques in form of articles, videos, and interactive modules on exercise, diet, and stress management for healthy heart lifestyle to manage sudden happening heart attack issues and heart related diseases.	8	Given that a user, When I access the HealthHub platform, Then there should be a prominently displayed section dedicated to exercise, diet, and stress management specifically tailored for heart health.
18	As a user entrusting my health data to HealthHub, I want assurance that my information is protected from unauthorized access and breaches so that I did not face any problem in future	9	Given that I am a user of HealthHub when I access the platform or submit any health-related information then all data transmission between my device and the HealthHub servers should be encrypted using industry-standard protocols

19	As a doctor, I want secure exchange of health data of my patients' health and provide them with up-to-date information on their health, so that they get information for their well-being.	9	Given that I am a healthcare provider registered with HealthHub, when I communicate with patients or access their health records through the platform, then all communication channels should be encrypted to, and patient data should only be accessible to authorized healthcare providers with proper authentication.
20	As a user of HealthHub, I want clear and transparent information about how my health data is handled and protected to make informed decisions about sharing my information on the platform	9	Given that I am a user of HealthHub, when I review the privacy policy and terms of service, then the policy should clearly outline how user data is collected, stored, and used

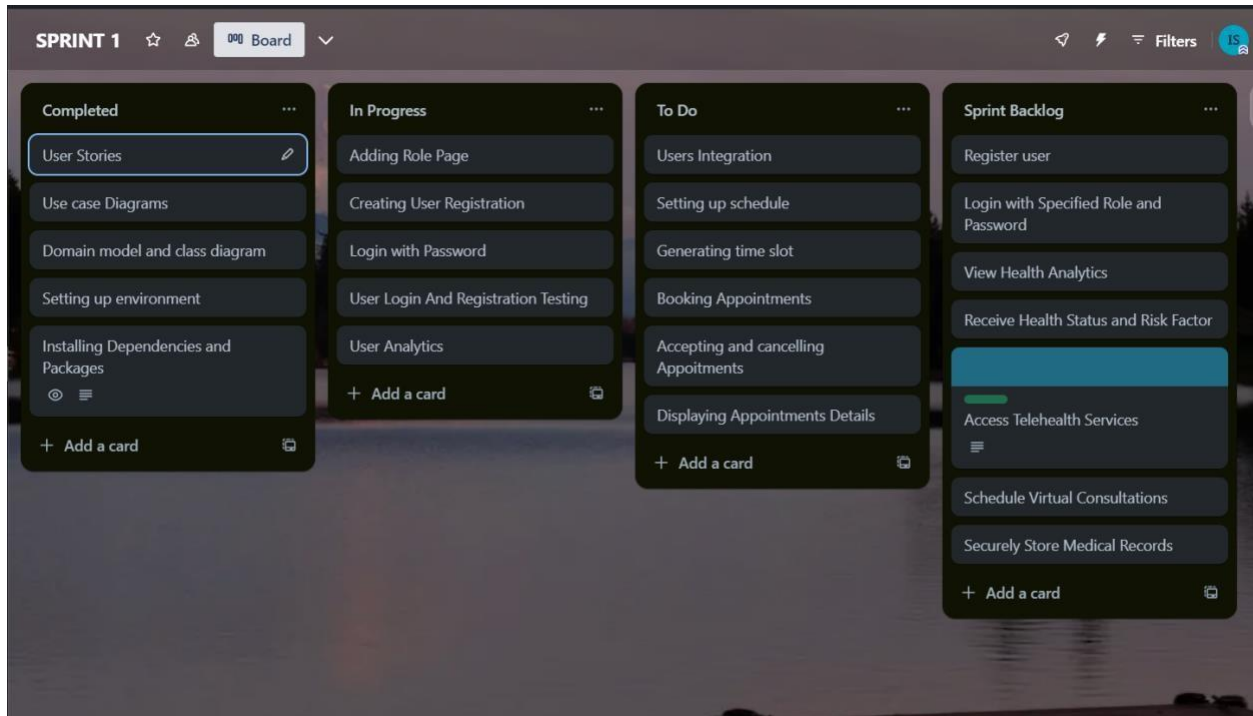
21	As an administrator of Health Hub, I want to monitor the activities of doctors, clients, and appointments so that I can efficiently manage the healthcare services provided through the platform.	1	Given that I am logged in as an administrator, when I navigate to the admin dashboard, I should see an overview of all registered doctors, clients, and upcoming appointments.
22	As a user, I want to receive personalized adherence plans for my medications within HealthHub so that it helps me stay organized and ensure I take them correctly.	10	Given that user has enabled the medical adherence feature when users input the information about their prescribed medications, including name, dosage, and instructions then it should give reminder at specified time
23	As a user, I want to receive timely reminders for my medication doses so that it helps me remember to take them on time.	10	Given that the user has provided the prescribed medications information when it is the specified time then system should give push notifications on the HealthHub app, email, or SMS, according to user preferences.
24	As a user, I want to get my real-time health insights while monitoring devices so that it collects and evaluate different activity levels.	11	Given that a remote monitoring device is connected to HealthHub when I start using it then it should update the user's health dashboard with the latest data received from each connected device.
25	As a user of HealthHub, I want the platform to analyze my health data to provide me with personalized insights into my health risks and challenges.	12	Given that I am a user of HealthHub, when I provide my health data such as medical history, lifestyle habits, and biometric measurements, Then the platform should utilize AI-driven predictive analytics to analyze my data and identify potential health risks

Product Gantt Chart:

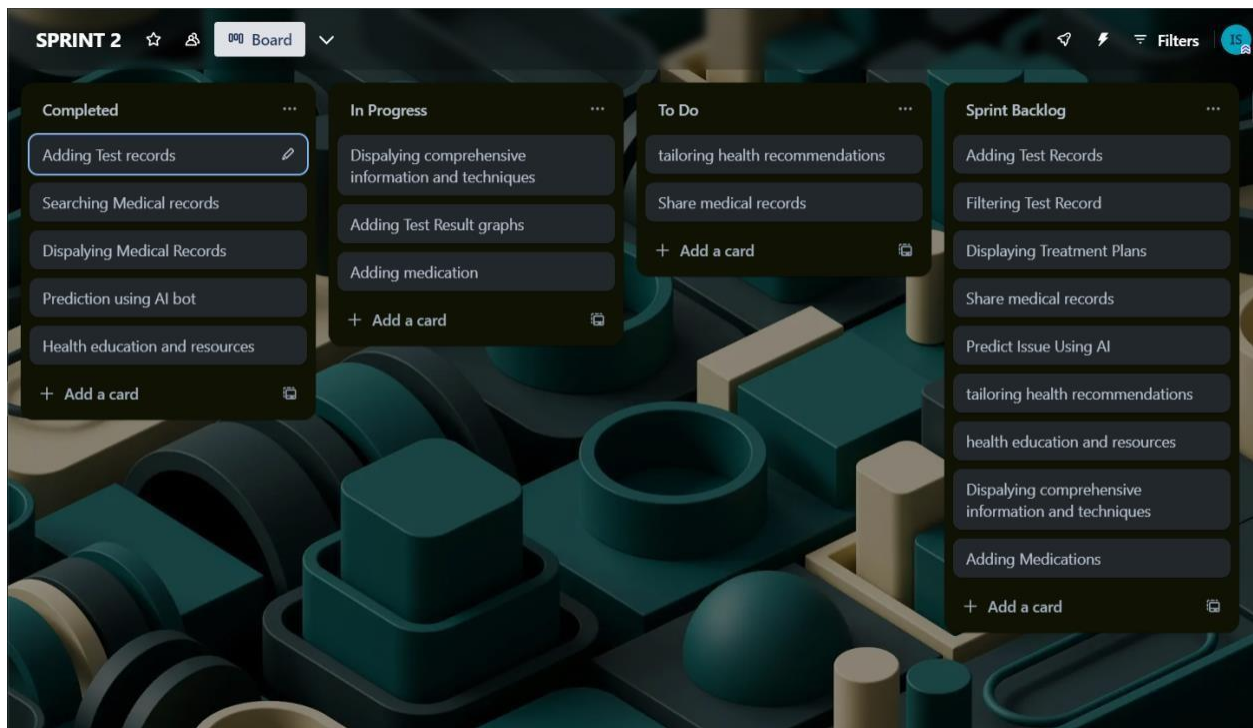


Trello:

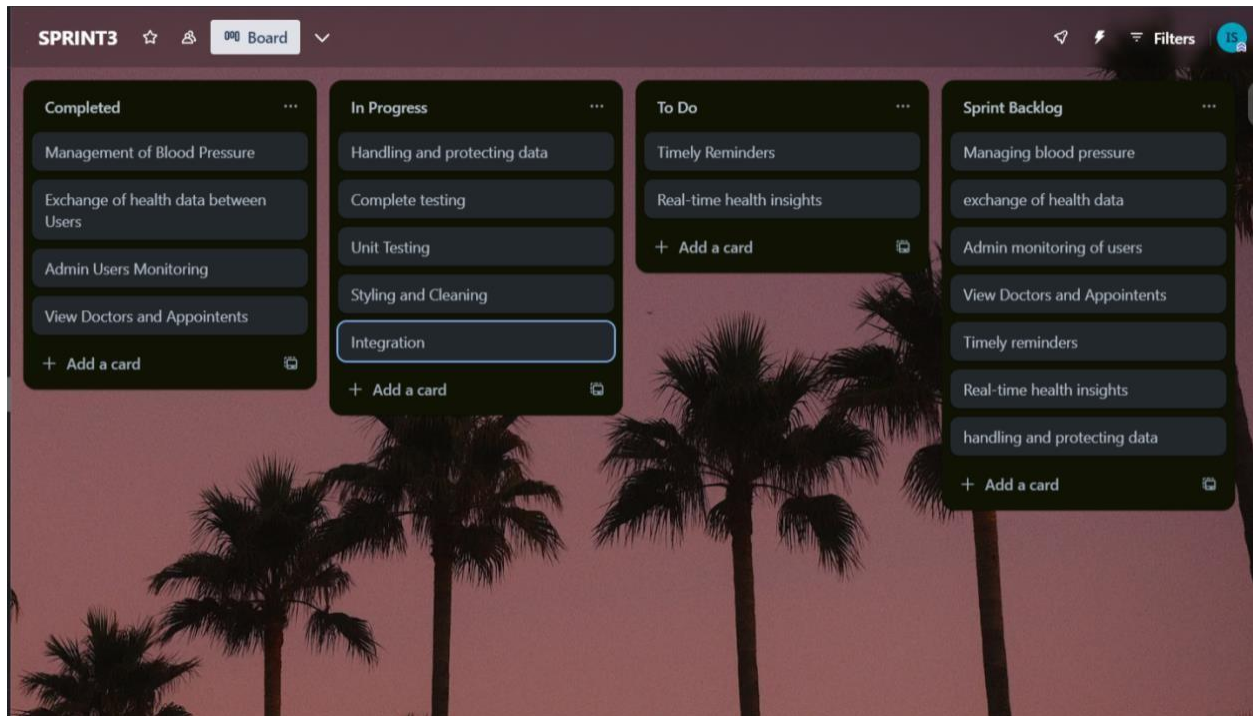
Sprint 1:



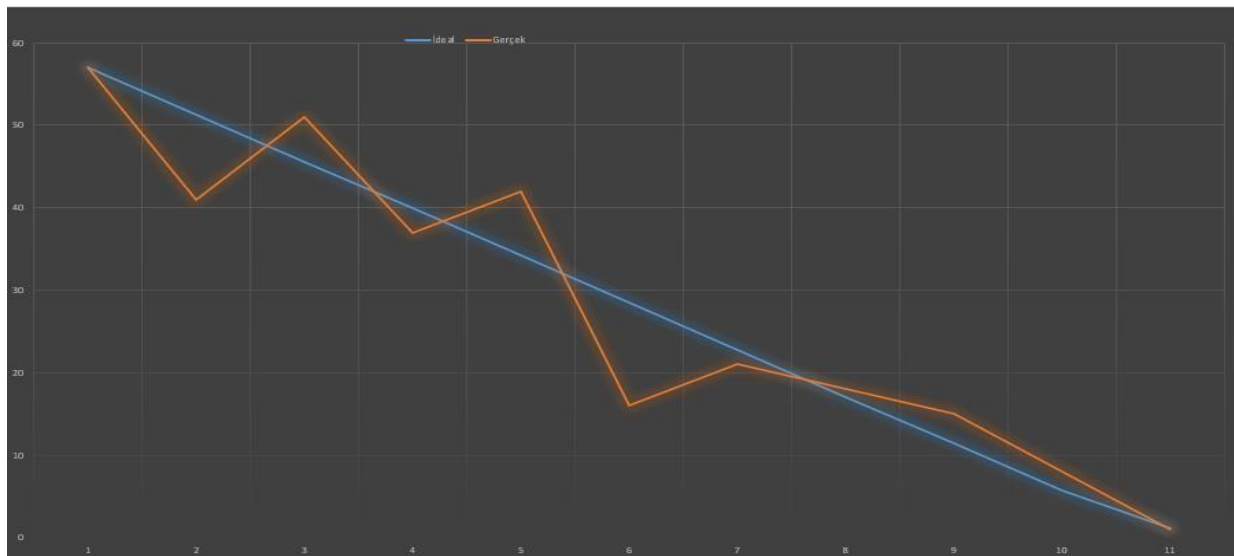
Sprint 2:



Sprint 3:

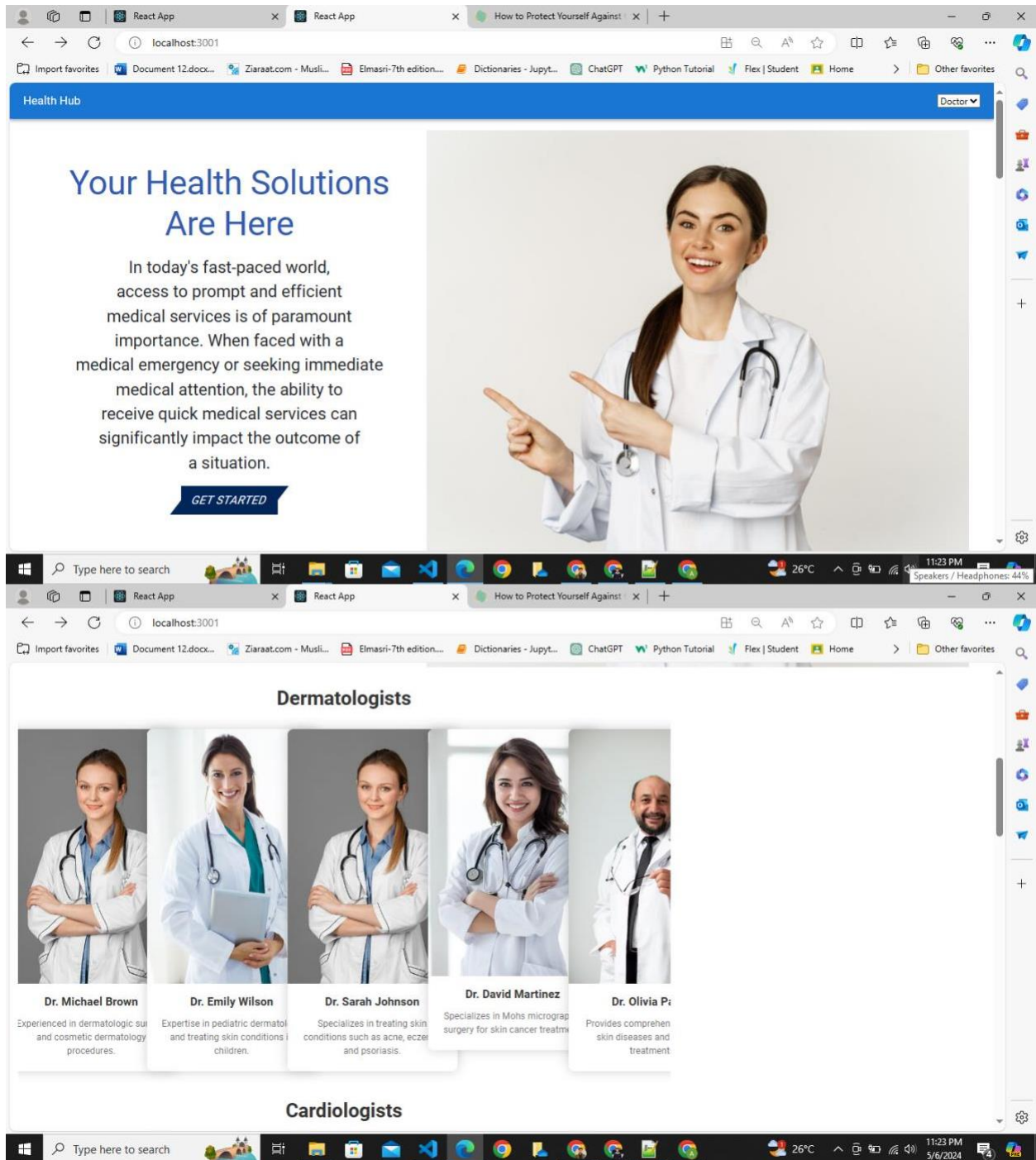


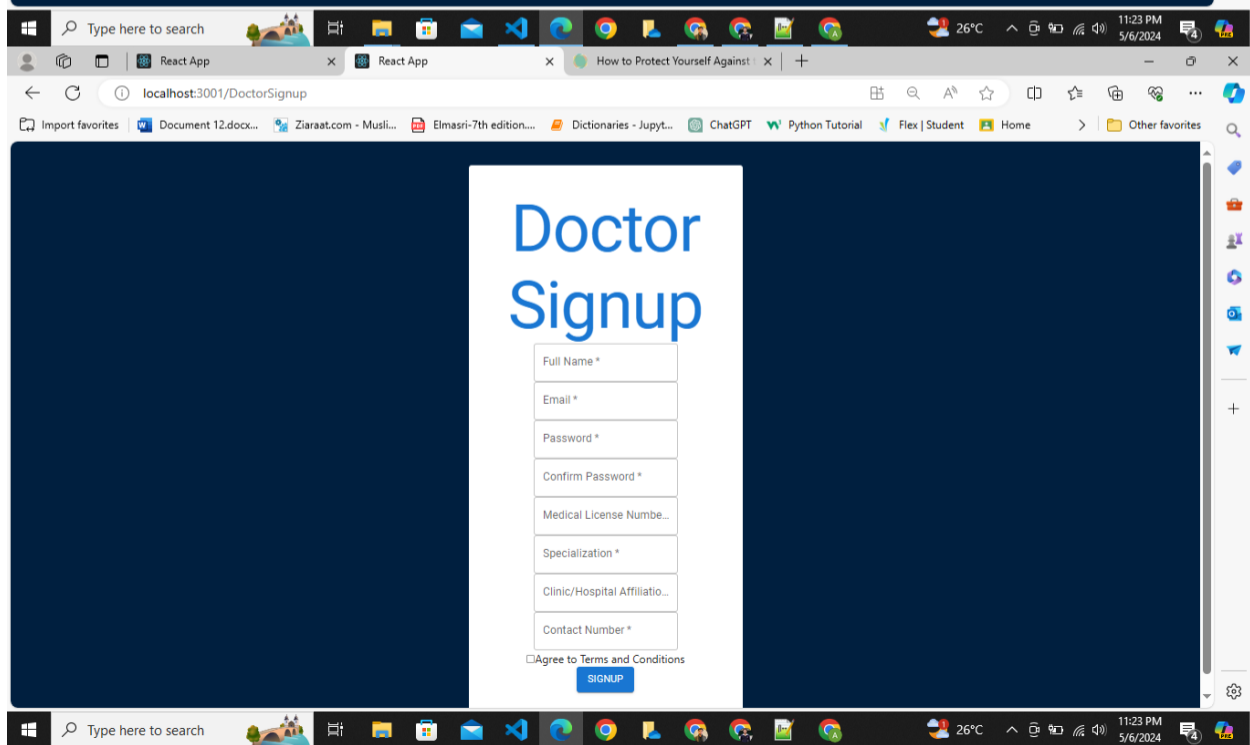
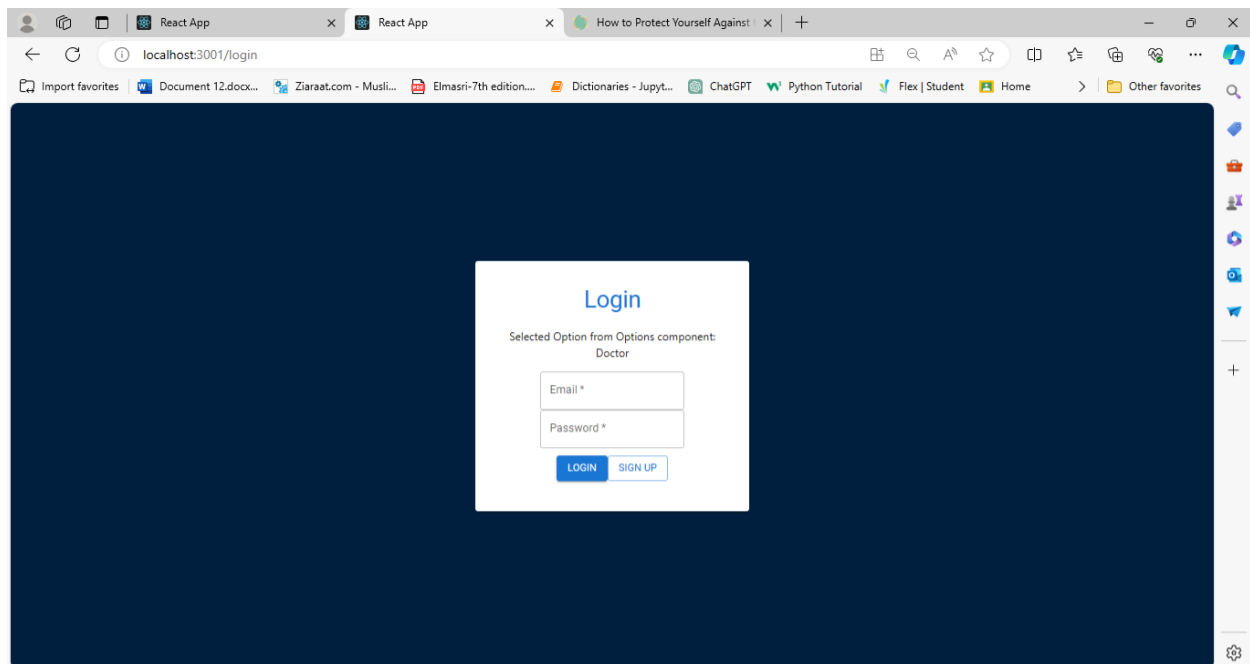
Product Burndown Chart:



Project GUI:

The Role Page of HealthHub presents users with three distinct buttons labeled "Client," "Doctor," and "Admin," enabling them to easily select their role within the system. Additionally, the page displays a comprehensive list of doctors, categorized by their specialties, providing users with convenient access to healthcare professionals relevant to their needs.





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localhost:3001/ClientSignup

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Client Signup

Full Name *

Email *

Password *

Confirm Password *

Birth dd / yyyy *  ▼

Medical History (option...

Contact Number *

☐ Agree to Terms and Conditions

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Admin Signup

Full Name *

Email *

Password *

Confirm Password *

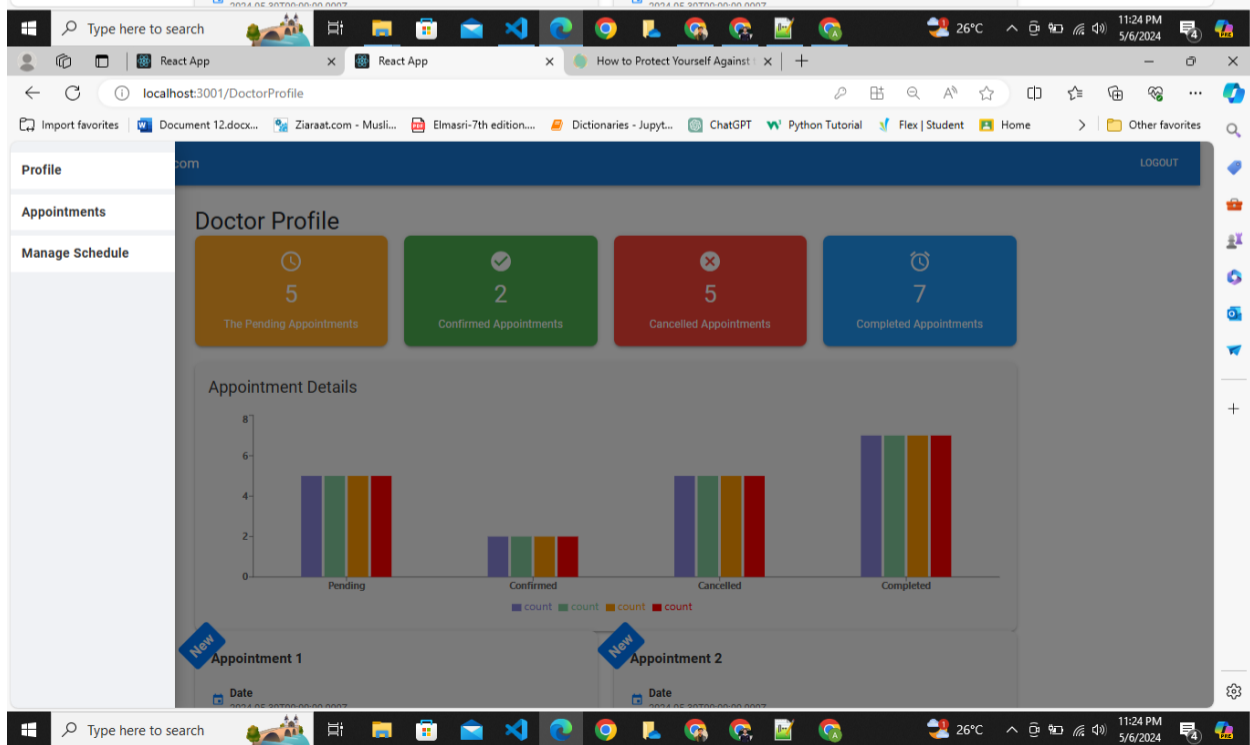
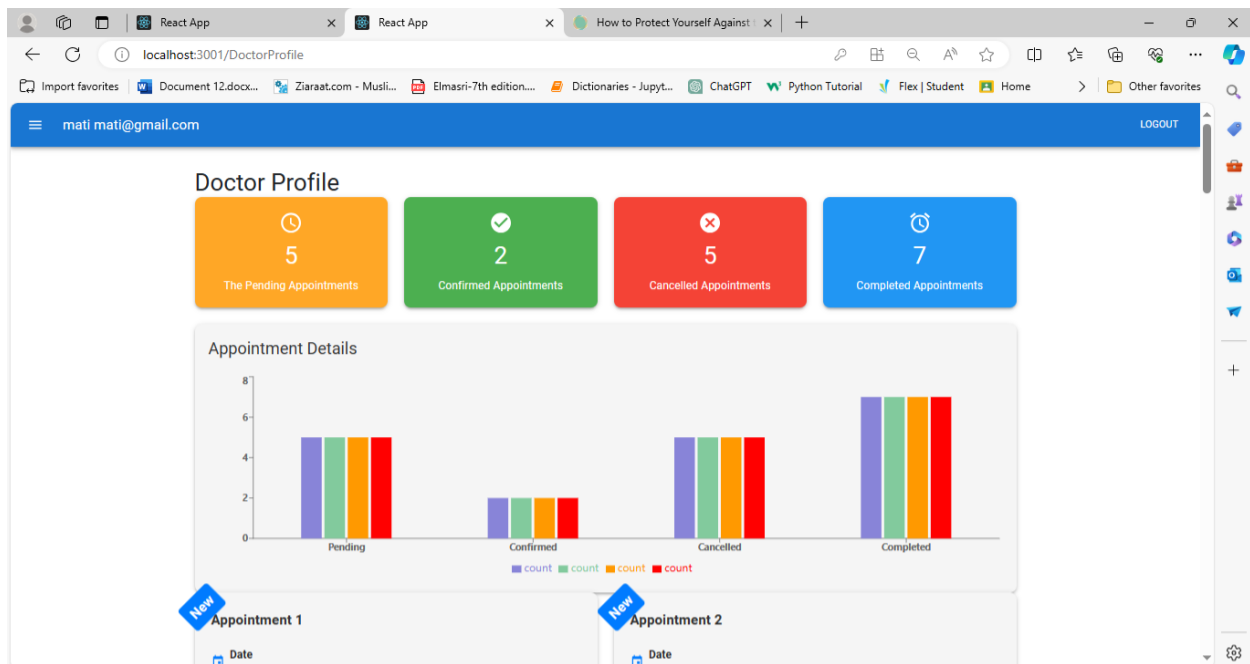
Role *

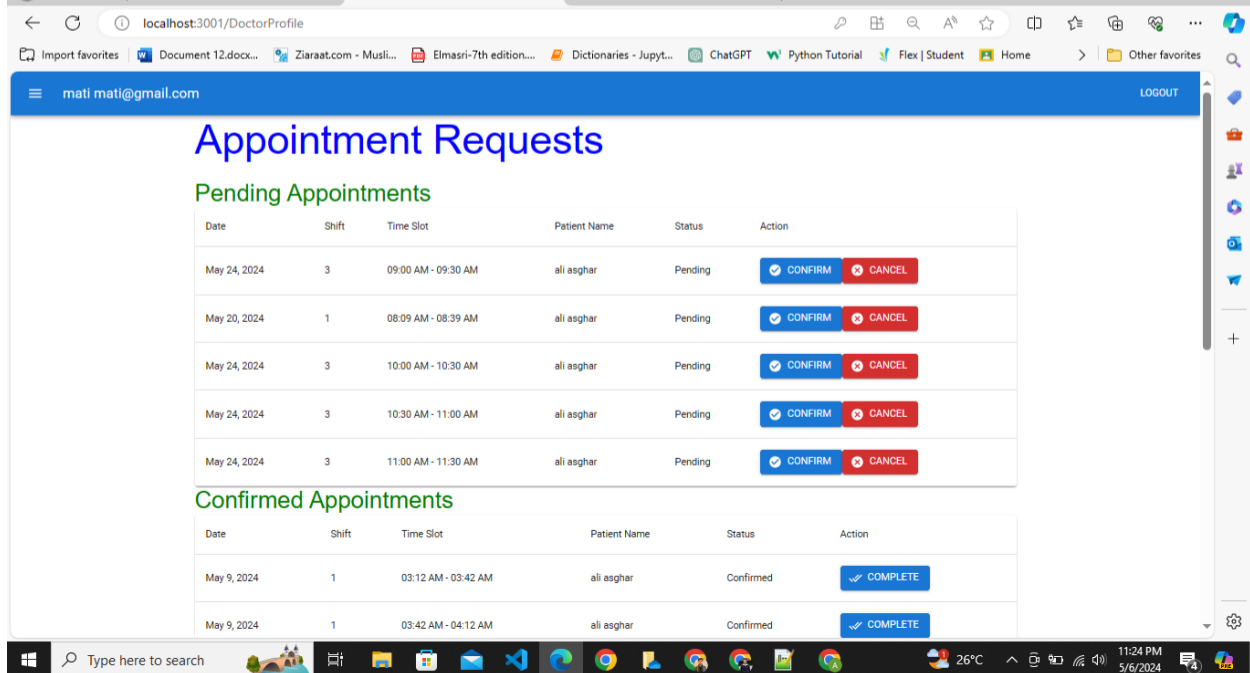
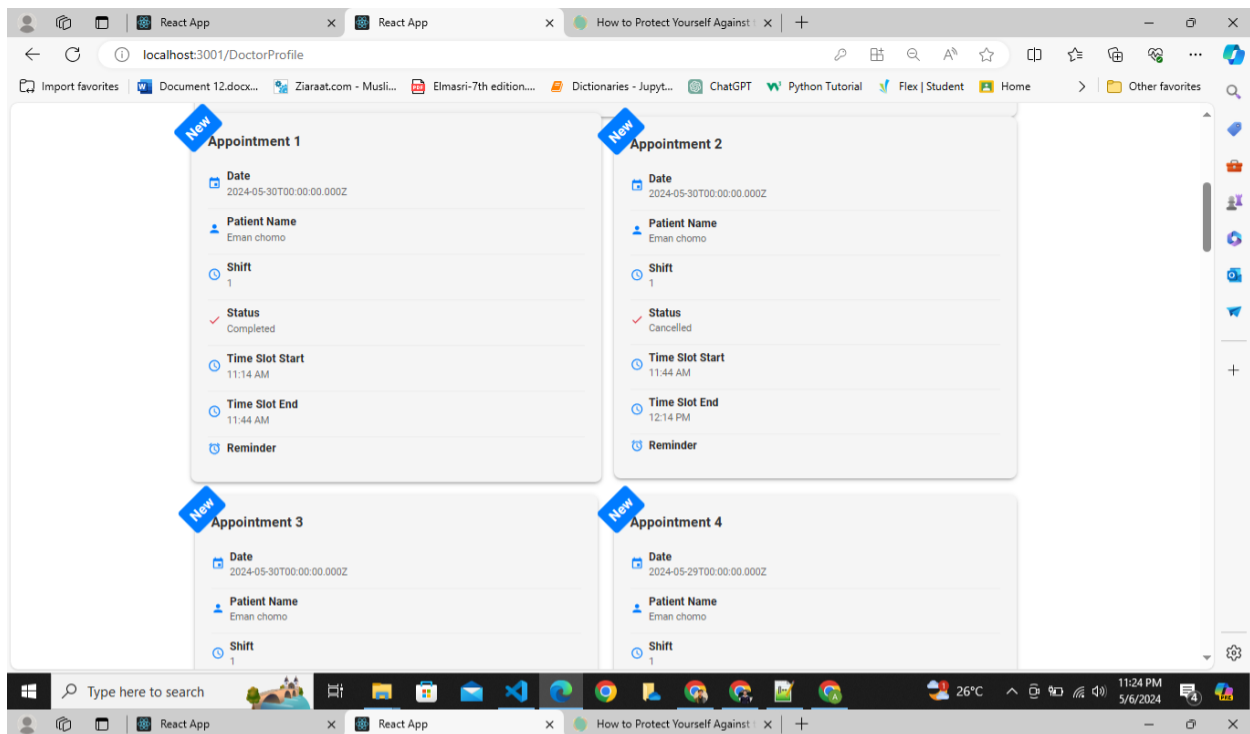
Admin

Contact Number *

☐ Agree to Terms and Conditions

SIGNUP





May 9, 2024	1	03:42 AM - 04:12 AM	ali asghar	Confirmed	COMPLETE
-------------	---	---------------------	------------	-----------	----------

Cancelled Appointments

Date	Shift	Time Slot	Patient Name	Status
May 30, 2024	1	11:44 AM - 12:14 PM	Eman chomo	Cancelled
May 29, 2024	1	11:44 AM - 12:14 PM	Eman chomo	Cancelled
May 20, 2024	1	07:39 AM - 08:09 AM	ali asghar	Cancelled
May 24, 2024	3	08:00 AM - 08:30 AM	ali asghar	Cancelled
May 24, 2024	3	09:30 AM - 10:00 AM	ali asghar	Cancelled

Completed Appointments

Date	Shift	Time Slot	Patient Name	Status
May 30, 2024	1	11:14 AM - 11:44 AM	Eman chomo	Completed
May 30, 2024	1	12:14 PM - 12:44 PM	Eman chomo	Completed
May 29, 2024	1	11:14 AM - 11:44 AM	Eman chomo	Completed
May 29, 2024	1	12:14 PM - 12:44 PM	ali asghar	Completed
May 20, 2024	1	07:09 AM - 07:39 AM	ali asghar	Completed
May 24, 2024	3	08:30 AM - 09:00 AM	ali asghar	Completed

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matl matl@gmail.com

LOGOUT

Manage Schedule

Date *
mm/dd/yyyy

Shift ID *

Start Time *
--:--

End Time *

ADD/UPDATE SCHEDULE

Schedule List

Date: 5/4/2024
Shift ID: 1
Start Time: 10:31
End Time: 04:01 PM

Date: 5/9/2024
Shift ID: 1
Start Time: 11:14
End Time: 05:14 PM

Date: 5/14/2024

Calendar

Today Back Next

May 2024

Month Week Day Agenda

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	01	02	03	04

localhost:3001/DoctorProfile

ADD/UPDATE SCHEDULE

Date: 5/9/2024

Shift ID: 1

Start Time: 11:14

End Time: 05:14 PM

Date: 5/14/2024

Calendar

Today Back Next

May 2024

Month Week Day Agenda

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	01	02	03	04 Shift ID: 1 +2 more
05	06 Shift ID: 1	07	08	09 Shift ID: 1 Shift ID: 1	10	11 Shift ID: 1
12	13	14 Shift ID: 1	15	16	17	18 Shift ID: 1
19	20 Shift ID: 1	21	22	23 Shift ID: 1	24 Shift ID: 3 Shift ID: 1	25 Shift ID: 1 Shift ID: 1
26 Shift ID: 1 Shift ID: 1	27 Shift ID: 1	28 Shift ID: 1	29 Shift ID: 2 Shift ID: 1	30 Shift ID: 1	31 Shift ID: 1	01

Admin Dashboard

TOGGLE ADMIN PROFILE

VIEW DOCTORS

Admin Profile

Full Name: Husnain Ali

Email: husnainali2721@gmail.com

Contact Number: 03484485412

Profile Creation Date: 16-09-2003

Last Updated: Currently

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localhost:3001/doctors

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Admin Dashboard

List of Doctors

This page will show lists of all the Registered Doctors and you can view their all record of appointments .

Ali

Email: alikhizar9110@gmail.com

Contact Number: a

Medical License Number: 1

Specialization:

Clinic/Hospital:

[VIEW MORE](#)

ali abbas

Email: aliabbas9110@gmail.com

Contact Number: a

Medical License Number: 1

Specialization:

Clinic/Hospital: a

[VIEW MORE](#)

ali

Email: ali110@gmail.com

Contact Number: 110

Medical License Number: ali

Specialization:

Clinic/Hospital: ali

[VIEW MORE](#)

ali

Email: ali514@gmail.com

Contact Number: 336

Medical License Number: 514

Specialization:

Clinic/Hospital:

[VIEW MORE](#)

Eman

Email: eman@gmail.com

Contact Number: 336

Medical License Number: 110

Specialization:

Clinic/Hospital:

[VIEW MORE](#)

talha

Email: talha@gmail.com

Contact Number: 69

Medical License Number:

Specialization:

Clinic/Hospital:

[VIEW MORE](#)

Type here to search

26°C

11:25 PM

5/6/2024

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localhost:3001/appointments/662e546889ecc9a4da3c0816

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Admin Dashboard

Doctor Details

Name: mati

Email: mati@gmail.com

Contact Number: mati

Medical License Number: mati

Specialization: mati

Clinic/Hospital: mati

Appointment Count: 19

Appointment Records

Date: 2024-05-30T00:00:00.000Z

Status: Completed

Date: 2024-05-30T00:00:00.000Z

Status: Cancelled

Date: 2024-05-30T00:00:00.000Z

Status: Completed

Date: 2024-05-29T00:00:00.000Z

Status: Completed

Date: 2024-05-29T00:00:00.000Z

Status: Cancelled

Date: 2024-05-29T00:00:00.000Z

Status: Completed

Date: 2024-05-20T00:00:00.000Z

Status: Completed

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26°C

11:25 PM

5/6/2024

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User ProfileLOGOUT

B+

Blood Type

Lung Health

55%

Age: 21 years

Height: 5 metres

Weight: 72 KGs

BMI: 2.88

Test Results

Add Test Results

Upper BP: 100

Lower BP: 120

Heart Rate: 45

Date: 2024-05-18

Upper BP: 100

Lower BP: 120

Heart Rate: 45

Date: 2024-05-11

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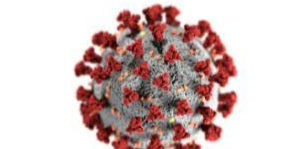


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User Profile

LOGOUT

B+

Blood Type

Lung Health

55%

Age: 21 years

Height: 5 metres

Weight: 72 KGs

BMI: 2.88

Test Results

Add Test Results

Upper Blood Pressure *

Lower Blood Pressure *

Heart Rate *

Date *
mm/dd/yyyy

SUBMIT

CLOSE

Type here to search

26°C

11:27 PM

5/6/2024

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User Profile

LOGOUT

Appointments

Doctors

Ali

Contact Number: a

Specialization: N/A

Medical License Number: 1

Clinic or Hospital: N/A

BOOK APPOINTMENT

ali abbas

Contact Number: a

Specialization: N/A

Medical License Number: 1

Clinic or Hospital: a

BOOK APPOINTMENT

ali

Contact Number: 110

Specialization: N/A

Medical License Number: ali

Clinic or Hospital: ali

BOOK APPOINTMENT

ali

Contact Number: 336

Specialization: N/A

Medical License Number: 514

Clinic or Hospital: N/A

BOOK APPOINTMENT

Eman

Contact Number: 336

Specialization: N/A

Medical License Number: 110

Clinic or Hospital: N/A

BOOK APPOINTMENT

mati

Contact Number: mati

Specialization: mati

Medical License Number: mati

Clinic or Hospital: mati

BOOK APPOINTMENT

Eman chomo

Contact Number: 69

Specialization: chomo sarkar

Medical License Number: 121

Clinic or Hospital: chomiyen centre

BOOK APPOINTMENT

tariq bhai

Contact Number: 12345

Specialization: mbbs

Medical License Number: 110

Clinic or Hospital: lodhran

BOOK APPOINTMENT

ali asghar

Contact Number: 12345

Specialization: business health :)

Medical License Number: 1

Clinic or Hospital: poly clinic

BOOK APPOINTMENT

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localhost:3001/ClientProfile

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User Profile

LOGOUT

Appointments

Book Appointment

Doctor: ali abbas

Doctorid: 6622e2751d0ea9d4d1e74816

Clientid: 6622ff2e4afe219c6c8ec35c

Select Date:

Select Date

Book Appointment

Cancel

Doctors

Ali

Contact Number: a

Specialization: N/A

Medical License Number: 1

Clinic or Hospital: N/A

BOOK APPOINTMENT

ali abbas

Contact Number: a

Specialization: N/A

Medical License Number: 1

Clinic or Hospital: a

BOOK APPOINTMENT

ali

Contact Number: 110

Specialization: N/A

Medical License Number: ali

Clinic or Hospital: ali

BOOK APPOINTMENT

ali

Contact Number: 336

Specialization: N/A

Medical License Number: 514

Eman

Contact Number: 336

Specialization: N/A

Medical License Number: 110

mati

Contact Number: mati

Specialization: mati

Medical License Number: mati

React App

React App

React App

+

localhost:3001/ClientProfile

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User Profile

LOGOUT

Search by doctor name or date

Status

Appointments

Appointment Status: Completed

Doctor: mati

Date: Wed May 29 2024

Start Time: 12:14 PM

End Time: 12:44 PM

Appointment Status: Completed

Doctor: mati

Date: Mon May 20 2024

Start Time: 07:09 AM

End Time: 07:39 AM

Appointment Status: Cancelled

Doctor: mati

Date: Mon May 20 2024

Start Time: 07:39 AM

End Time: 08:09 AM

React App x React App x React App x

localhost:3001/ClientProfile

User Profile LOGOUT

Search by doctor name or date

Appointments

Appointment Status: Completed

Doctor: mati

Date: Wed May 29 2024

Start Time: 12:14 PM

End Time: 12:44 PM

Appointment Status: Completed

Doctor: mati

Date: Mon May 20 2024

Start Time: 07:09 AM

End Time: 07:39 AM

Appointment Status: Cancelled

Doctor: mati

Date: Mon May 20 2024

Start Time: 07:39 AM

End Time: 08:09 AM

Status

All

Confirmed

Pending

Cancelled

Completed

Type here to search

26°C

11:27 PM

5/6/2024

React App x React App x React App x

localhost:3001/ClientProfile

User Profile LOGOUT

Medical Records

Search by doctor name, date, diagnosis, or prescription...

SEARCH

Doctor: Ali

Date: 4/6/2024

Diagnosis: Asthma

Prescription: Bruffen

Doctor: akbar

Doctor: talha

Doctor: talhabutt

Doctor: butt

Doctor: butt

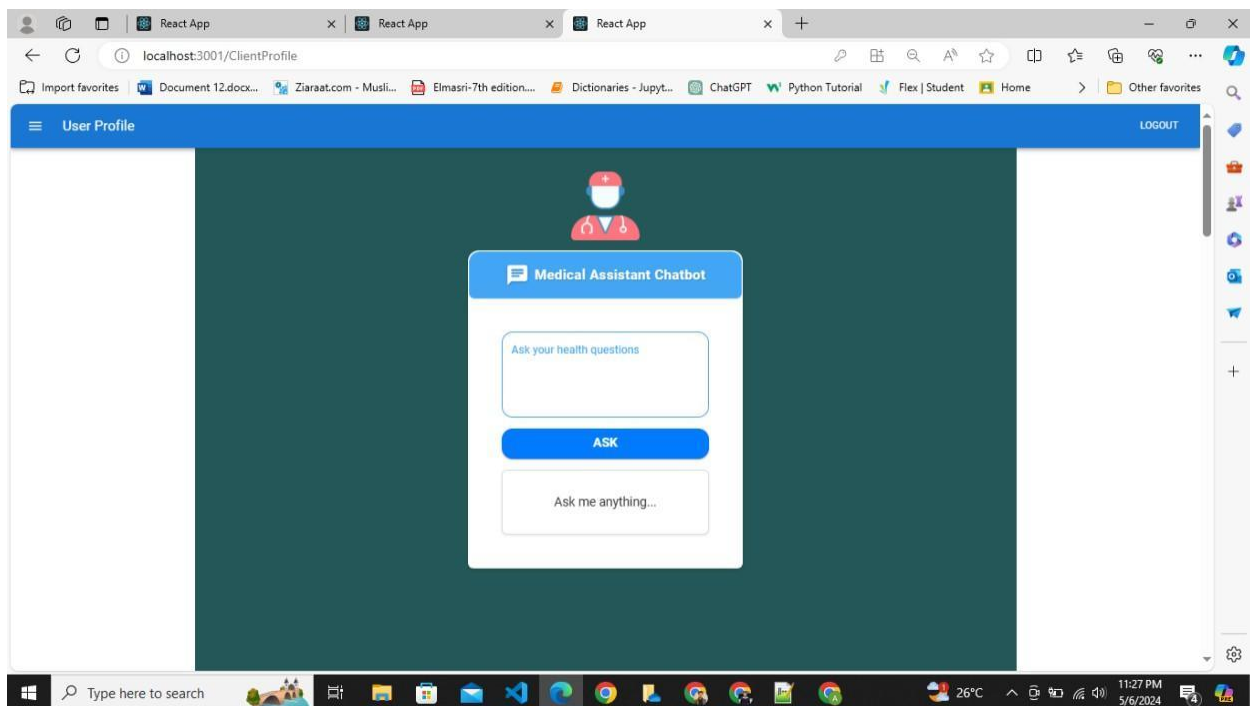
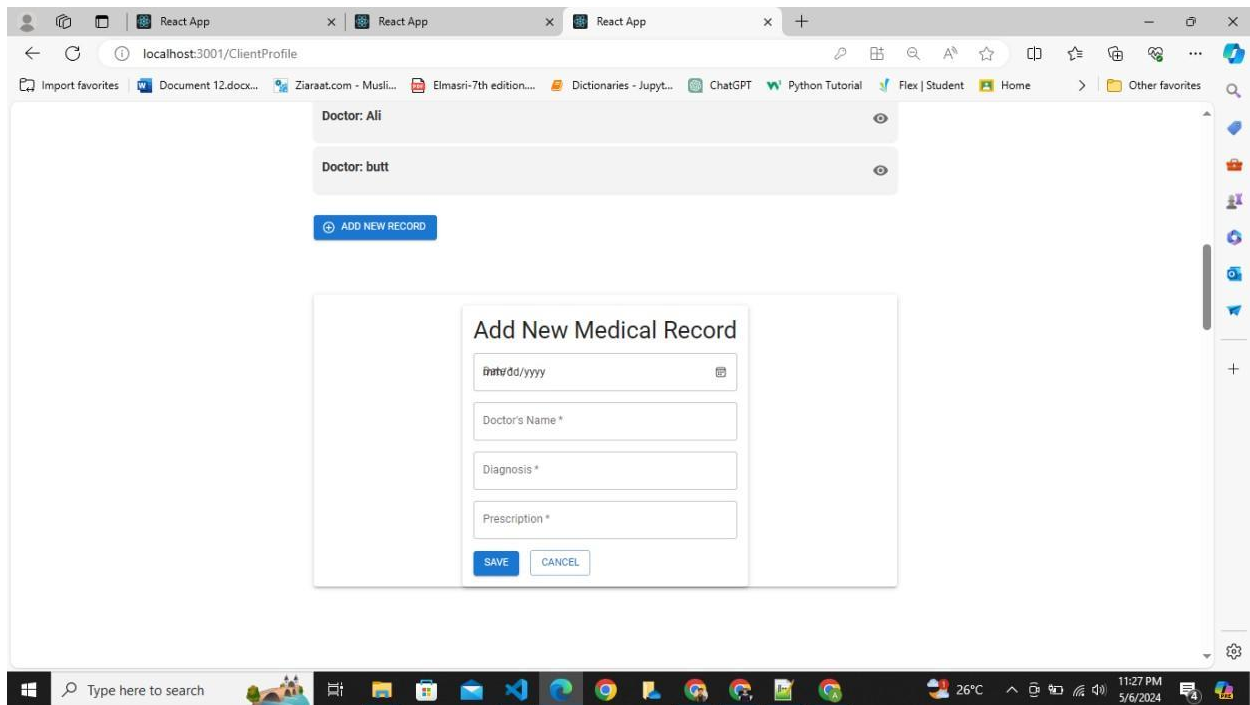
Doctor: butt

Type here to search

26°C

11:27 PM

5/6/2024



Equivalent Class Partitioning and Boundary Values:

User Story 1: Registration

Equivalent Class Partitions:

Input or Output Event	Valid Equivalence Classes	Invalid Equivalence Classes
<i>Email field:</i> The User must enter their Email completely.	@ is a must .com, .org, .ai, are must	No @ or .com provided
<i>Password:</i> The User must provide a password so that they securely access their account.	An upper-case alphabet, a lower-case alphabet, a non-letter character, 6 chars<= Length <= 10 chars (The valid password registered in the database)	No Upper Case Alphabet No Lower-Case Alphabet No non-letter characters Length < 6 chars Length > 10 chars (anything other than the password registered in the database)
<i>Confirm Password:</i> The User must provide a password so that they can confirm their password.	Both password and Confirm password must be same.	Confirm Password and Password is not same
<i>Contact Number:</i> The User must provide their cell phone number for communication	Contact number must in n numeric form such as 1,2,3 and so on It must have 11 digits	It do not have 11 digits (Numeric validation are completed no one can add other than numeric data)

Boundary Value Analysis:

Input	Boundary Value
Email	Minimum valid length of an email address (e.g., 6 characters)
Password	Length of Password=6 and Username=10 are the boundary values and should be accepted
<i>Confirm Password:</i> The User must provide a password so that they can confirm their password.	Must be Same as password field value

Contact Number:	Contact number <=11
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User Story 2: Login

Equivalent Class Partitions:

Input or Output Event	Valid Equivalence Classes	Invalid Equivalence Classes
<i>Email field:</i> The User must enter their Email completely.	@ is a must .com, .org, .ai, are must	No @ or .com provided
<i>Password:</i> The User must provide a password so that they securely access their account.	An upper-case alphabet, a lower-case alphabet, a non-letter character, 6 chars<= Length <= 10 chars (The valid password registered in the database)	No Upper Case Alphabet No Lower-Case Alphabet No non-letter characters Length < 6 chars Length > 10 chars (anything other than the password registered in the database)

Boundary Value Analysis:

Input	Boundary Value
Email	Minimum valid length of an email address (e.g., 6 characters)
Password	Length of Password=6 and Username=10 are the boundary values and should be accepted

User Story 3: Personalized Health Information

Equivalent Class Partitions:

Input or Output Event	Valid Equivalence Classes	Invalid Equivalence Classes
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<i>Age:</i> User must add his age foe basic info to calculate BMI	Must be a number Must be positive Number	Number out of range Must not positive Number
<i>Height:</i> User must add his age foe basic info to calculate BMI	Must be a number Must be positive Number	Number out of range Must not positive Number
<i>Weight:</i> User must add his age foe basic info to calculate BMI	Must be a number Must be positive Number	Number out of range Must not positive Number

Boundary Value Analysis:

Input	Boundary Value
<i>Age:</i>	Length of <i>Age</i> <=3 <i>Age</i> >0
<i>Height:</i>	Length of <i>Height</i> <=1 <i>Height</i> >0
<i>Weight:</i>	Length of <i>Height</i> <=3 <i>Weight</i> >0

User Story 4: Test results

Equivalent Class Partitions:

Input or Output Event	Valid Equivalence Classes	Invalid Equivalence Classes
Date field: User must enter date to add test record of that date	Accurate day, month and the year is chosen according to the format	Number out of range Must not positive Number
<i>Upper Blood Pressure:</i> User must enter Blood Pressure to add a test result	Must be a number Must be positive	Number out of range Must not positive Number

<i>Lower Blood Pressure:</i> User must enter Blood Pressure to add a test result	Must be a number Must be positive	Number out of range Must not positive Number
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<i>Heartbeat rate:</i> User must enter Heart beat rate to add a test result	Must be a number Must be positive	Number out of range Must not positive Number
--	--------------------------------------	--

Boundary Value Analysis:

Input	Boundary Value
Date	Must be less than current date
Upper Blood Pressure	Length of Upper Blood Pressure ≤ 3 Upper Blood Pressure > 0
Lower Blood Pressure	Length of Lower Blood Pressure ≤ 3 Lower Blood Pressure > 0
Heartbeat rate	Length of Heartbeat rate ≤ 3 Heartbeat rate > 0

User Story 5: Health Analytics

Equivalent Class Partitions User Story 5:

Input or Output Event	Valid Equivalence Classes	Invalid Equivalence Classes
<i>Health Data estimation:</i> User must click the button provided to check for health data calculation	Clicked only once for estimation Clicked again for results	Clicked more than once or not at all
<i>Blood type:</i> User must choose the blood type to store the information.	Click to add data Must choose a value from drop down	Do not clicked Didn't selected an option

No Boundary Value Analysis is possible for User Story 5.

User Story 6: Appointments Management

Equivalent Class Partitions User Story 6:

Input or Output Event	Valid Equivalence Classes	Invalid Equivalence Classes
<i>Search for Appointments:</i> User must add Doctor name or specification to get for a specific doctor slot	Must add alphabetic Data Doctor must be part of System	Data entered other than alphabetic User not part of system
<i>Select timeslot:</i> User must choose a valid timeslot of doctor to ask for appointment acceptance	Must Select a valid Slot from dropdown data	Did not select a valid slot Did not selected anything
<i>Time slots generation:</i> User must enter a valid date for generating slots	Must chose a valid date	Date not selected Did not selected a valid date
<i>Accepting Appointment:</i> The Doctor must choose the accepted or cancelled to change the status of appointment	Must click on the button provided	Did not clicked the button provided

Boundary Value Analysis:

Input	Boundary Value
Search for Appointments	Must be in alphabetic form. Any text with length $\geq$ 0
Select timeslot date	Must be in range Year and month must be current

Improvements:

1. **Enhanced User Experience:** Streamlining the user interface further could improve overall user experience, making navigation more intuitive and efficient.
2. **Personalization:** Implementing more personalized features based on user preferences and behavior could enhance engagement and satisfaction.
3. **Expand Feature Set:** Continuously adding new features such as integration with wearable devices for real-time health monitoring or incorporating AI-driven recommendations could enrich the platform's offerings.
4. **Accessibility:** Ensuring accessibility for users with disabilities by implementing features such as screen reader compatibility and keyboard navigation would broaden the platform's reach.
5. **Performance Optimization:** Optimizing system performance, particularly in areas like load times and responsiveness, can further enhance user satisfaction.

Limitations:

1. **Data Security Concerns:** Ensuring robust data security measures to safeguard sensitive user information is crucial to maintain trust and compliance with data protection regulations.
2. **Technical Challenges:** Addressing technical challenges such as system scalability and compatibility across different devices and browsers is essential for a seamless user experience.
3. **Resource Constraints:** Limited resources, including budget and manpower, may impact the speed and scope of project development and implementation.
4. **User Adoption:** Encouraging user adoption and engagement, particularly among healthcare professionals and clients, may require targeted marketing efforts and user education initiatives.

GitHub repository Link: